



Callum Mackenzie

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EXPERIENCE

Software Development Engineer Intern

May 2026 – Present

Amazon

Vancouver, BC, Canada

- Developing an **agentic AI platform on AWS** that automates **ticket root-cause analysis** and continuously expands its capabilities through operational feedback loops
- Delivered **RAG**, source attribution, configurable **Bedrock KB integration**, auto-improving **CTI suggestions**, learning commands, and historical ticket knowledge management to **5 teams**
- Implementing **MCP/A2A config**, Slack actions, auto-execution, **metrics dashboard**, and model fine-tuning

Embedded Software Engineer Co-op

Jan 2025 – May 2026

General Dynamics

Calgary, AB, Canada / Remote

- Designed software for **tactical communications systems** on a **defense R&D team**
- Developed **3 Linux device drivers and daemons** in C/C++ for distributed embedded platforms
- Saved customers **\$20k+ per vehicle platform** by enabling additional users on existing hardware resources
- Implemented **configurable networking functionality** for distributed systems using layer 3/4/5 protocols
- **Resolved 30+ defects** in embedded projects, improving reliability in safety-critical environments
- **Reduced CI/CD pipeline runtime by 50%** across multiple pipelines with Docker, Python, and Jenkins
- Built and **integrated AI infrastructure and tooling** supporting **20+ engineers** and 3 build pipelines
- Shipped **client functionality in .NET apps** interfacing with embedded devices & AI services

EDUCATION

The University of British Columbia

Vancouver, BC, Canada

BSc Computer Science & Statistics (GPA 4.0/4.33)

Sep 2022 – May 2027

Coursework: Computer Graphics, Software Engineering, Operating Systems, Algorithm Design & Analysis, Programming Languages, Statistical Modelling, Stochastic Processes, Applied Linear Algebra

PROJECTS

FITNET - Embedded Motion Sensing with EMG | C, Swift, iOS, FreeRTOS, Fusion360

Feb – Dec 2025

- Created wearable system ground-up to classify body movement patterns in real time (walking, running, etc)
- **Motion-tracked & analyzed body position** with custom wearables measuring EMG and IMU data
- Predicted movement category with **dynamic time warping KNN classification** with 98% accuracy
- Created **iOS app** with SwiftUI & SceneKit to **read & process device data** with optimized BLE
- Integrated microprocessor firmware with **Bluetooth Low-Energy** and iPhone CoreBluetooth for communication

Deloitte ThinkTech | React, Typescript, Javascript, Node, R, DocuSign, UIPath

Sep – Nov 2022

- Developed an **MVP for a real-world client business problem** regarding document tracking in the healthcare industry with two business and one technical teammate, placing **4th out of 160+ candidates**
- **Presented** solution overview for **100+ Deloitte employees & executives**, fellow competitors, and visitors
- Used UIPath RPA, DocuSign, and AWS, focusing on security to **manage sensitive records for client** company

Exvi Fitness | Java, Kotlin, AWS, R, Android, Git

Dec 2021 - Jun 2022

- Developed a **full-stack desktop & Android application** for tracking personal workouts and sharing with others
- **Webscrapped and cleaned the data of 2000+ exercises** from online datasets using Java and R
- Built **custom authentication system with 2FA and cryptographic storage** for sensitive user data
- **Synced across devices with a serverless backend** (AWS), allowing sharing workouts with friends
- Followed **responsive UI design principles**, allowing multi-platform capability and intuitive use with Kotlin

TECHNICAL SKILLS

Languages: Python, Java, C, C++, C#, Rust, TypeScript, JavaScript, Kotlin, Swift, R, GLSL

Frameworks & Libraries: .NET, Node.js, SwiftUI, OpenGL, FreeRTOS, ESP-IDF, Numpy, Pandas

Cloud & AI: AWS, AWS CDK, Amazon Bedrock, MCP, RAG, LLM Fine-Tuning, CNNs, Firebase

Build & Developer Tools: Git, Docker, Jenkins, Maven, CMake, gdb, Bash